Confidence Weighted Staged Supervision for Knee MRI Classification with Scarce Labels

Aadit P[1], Amrit Gopinath[1] and Adithya S[1]

[1]*Sri Sivasubramaniya Nadar College of Engineering, Chennai*

Abstract

Reading a knee MRI study means combining evidence from several series, planes and contrasts, and doing it for many possible findings at once. The 2026 RSNA Knee Abnormality Detection challenge asks for twelve such findings per study, but only 58 of its 4,407 training studies carry expert labels; the rest come with a free text radiology report written in one of many languages. We propose Confidence Weighted Staged Supervision (CWSS), in which labels extracted from reports by large language models are converted into confidence weights and used to pretrain a plane gated multiple instance network on 2.5D slice triplets, after which the network is refined on expert labels alone. We pair this with Anchored Delta Adaptation (ADA), a conservative way of adapting a strong public 384 pixel model to the expert labels while keeping its original weights as a fallback, and fuse the two branches. Staging alone raised our single model from 0.637 to 0.794 macro ROC AUC on the hidden test set, and the fused system reached 0.925. A variation study over twenty scored models shows that when weak supervision is applied matters as much as how much of it there is, and that 5 fold validation on 58 studies ranks models in close agreement with the hidden test set (Spearman $\rho = 0.95$).

Keywords

knee MRI, multilabel classification, weak supervision, large language models, multiple instance learning, transfer learning, radiology reports

I. Introduction

Knee MRI is one of the most frequently requested musculoskeletal examinations, and a single study can show several problems at the same time: a torn ligament, a meniscal tear, cartilage loss, fluid in the joint, bone bruising. A radiologist reads these across a set of series acquired in different planes and with different contrasts, then writes a free text report. Deep learning has shown that individual findings such as anterior cruciate ligament (ACL) tears or meniscal tears can be detected from MRI with performance close to that of radiologists [1], [3], [6]. Most of that work, however, depends on large cohorts in which every study has been labelled by experts.

The 2026 RSNA Knee Abnormality Detection challenge removes that comfort. The organisers released more than five thousand knee MRI examinations collected at sixteen sites on five continents, with reports written in about a dozen languages [30]. Twelve findings must be predicted for every study. In the training data we received, 4,407 studies have a report, but only 58 of them, about 1.3 percent, have the twelve expert labels. Everything else has to be learned from the reports, from the images themselves, or from models trained elsewhere.

Conventional fully supervised training is poorly suited to this setting: about 46 expert studies per training fold are too few to fit twelve findings across heterogeneous multisite scans, and our own expert only baselines confirm it. The reports are the natural alternative, because they describe what the radiologist saw in every study. Extracting labels from them is not enough on its own, however. Report labels are noisy, sometimes hedged, and silent about findings that were not mentioned, and a few dozen trusted expert labels must still be reconciled with thousands of weaker ones. The central question is therefore how report derived supervision should be ordered and weighted against expert labels when an image model is trained.

We entered as Team Bitzkrieg and used the competition as a controlled setting for that question. Over twenty scored models we moved from report classifiers trained on 46 reports, through semisupervised vision language models, to staged training with labels read by large language models (LLMs), and finally to a fused system whose best hidden test score is 0.925 macro ROC AUC.

This paper makes the following contributions.

1) **Confidence Weighted Staged Supervision (CWSS).** Confidence weighted LLM report labels are used for weak only pretraining, then discarded while the model is refined on expert labels alone; with identical labels this beat joint training (0.783 against 0.745).
2) **A plane gated top k multiple instance head** for bags of 2.5D slice triplets from up to six series.
3) **Anchored Delta Adaptation (ADA)** of a strong public prior, which tunes only its last stage and head and always keeps the original as a candidate.
4) **A heterogeneous resolution fusion** of the two branches, reaching 0.925.
5) **A variation study** across twenty models, including how far a 58 study cross validation can be trusted.

Section II reviews related work and the research gap, Section III the data and system, Sections IV and V the results and variations, and Section VI the limitations.

II. Related Work

A. Deep learning for knee MRI

MRNet [1] was one of the first systems to classify whole knee MRI examinations. It passed every slice of a series through a 2D network, max pooled the slice features and combined the three planes with logistic regression, reaching strong results for ACL and meniscal tears on 1,370 examinations from one institution. Liu *et al.* built segmentation and classification networks for cartilage lesions [2] and a fully automated pipeline for ACL tears [3]. Pedoia *et al.* used 3D networks to detect and stage meniscus and patellofemoral cartilage changes [4], and Roblot *et al.* and Fritz *et al.* reported meniscal tear detection compared against radiologists and surgery [5], [6]. Astuto *et al.* extended the scope to grading several abnormality types with a multitask framework [7]. Qiu *et al.* used co plane attention across sequences to diagnose twelve knee abnormalities on 1,748 subjects (mean AUC 0.812) [8], and Štajduhar *et al.* released the KneeMRI dataset for ACL injury detection [9].

Two observations matter for us. First, nearly all of these studies train on cohorts in which every case is expert labelled, often thousands of them. Second, Germann *et al.* found that a network trained on one homogeneous protocol lost performance on heterogeneous cohorts with different pulse sequences and field strengths [10], which is exactly the multisite situation of the RSNA data.

B. Labels from radiology reports

Because expert annotation is expensive, several groups have derived labels from reports. Wood *et al.* trained a report classifier on neuroradiology reports and used it to label head MRI datasets for computer vision [11]. With the arrival of LLMs, Mukherjee *et al.* showed that a locally run, privacy preserving model could label chest radiograph reports with agreement comparable to dedicated labelers [12]. These studies stop at the label: they measure how well reports are read, not how the resulting noisy labels should be fed to an image model, and none of them concern musculoskeletal MRI or multilingual reports.

C. Weak, noisy and multiple instance supervision

Zhou describes incomplete, inexact and inaccurate supervision as the three forms of weak labels [13]; report derived labels are both inexact (study level, not slice level) and inaccurate. Cheplygina *et al.* surveyed semisupervised, multiple instance and transfer learning in medical imaging [14], and Quellec *et al.* reviewed multiple instance learning for medical images and video [15]. Campanella *et al.* showed that slide level labels can train clinical grade pathology models when a study is treated as a bag of instances [16]. Karimi *et al.* reviewed remedies for noisy labels in medical image analysis, including loss weighting by label confidence [17].

D. Transfer, pretrained priors and ensembles

Tajbakhsh *et al.* found that fine tuning pretrained networks is at least as good as training from scratch in medical imaging and more robust to small training sets [18]. RadImageNet showed that pretraining on radiology images can help downstream medical tasks [19]. Ensembles of deep networks are a reliable way to reduce variance, as reviewed by Ganaie *et al.* [20].

E. Evaluating models with tiny test sets

Varoquaux and Cheplygina warn that small evaluation sets in medical imaging produce large uncertainty and can mislead model selection [21]. Maier-Hein *et al.* showed that rankings in biomedical image analysis challenges can change with small changes in test data or metric [22].

F. Research gap

Putting these strands together, we see four open problems.

- **G1** Knee MRI classifiers [1], [7], [8] assume large, fully annotated and mostly single site cohorts. Learning twelve findings when about one percent of studies are expert labelled has not been studied.
- **G2** Report and LLM labelling [11], [12] has been evaluated as a reading task. How such labels should be scheduled against a handful of expert labels when training an image model, jointly or in stages, and how their uncertainty should be weighted, remains open, and has not been examined for multilingual musculoskeletal reports.
- **G3** Transfer learning studies [18], [19] start from generic pretrained backbones. Adapting an already strong, task specific public model to a tiny expert set without destroying what it knows is not addressed.
- **G4** Evaluation literature [21], [22] cautions against small test sets, but whether a cross validation on a few dozen gold studies can still rank weakly supervised models has not been measured against an independent hidden test set.

These gaps lead to four research questions. RQ1: does staging weak and expert supervision beat mixing them? RQ2: does confidence weighting of LLM labels and the choice of LLM matter? RQ3: can a strong public prior be adapted to 58 studies safely? RQ4: how far can a 58 study cross validation be trusted? Taken together, the literature establishes that knee findings can be learned from fully labelled MRI and that reports can be read automatically, but it has not directly examined how noisy report derived supervision should be incorporated into image based multilabel knee MRI learning when expert labels are extremely scarce. Section III follows the order in which our evidence accumulated, describing each baseline together with the limitation that motivated the next step, and Sections IV and V answer RQ1 to RQ4 with controlled comparisons.

III. Methodology

A. Problem formulation

A study X is a set of series $\{S^{(1)}, \ldots, S^{(m)}\}$, and each series is an ordered stack of 2D slices with a plane $\pi \in$

$\{\text{sagittal}, \text{coronal}, \text{axial}\}$. For the twelve findings $c = 1, \ldots, 12$ the model outputs

$$\hat{p}_c(X) = \sigma\big(s_c(X)\big), \qquad \hat{p}_c \approx P(y_c = 1 \mid X), \tag{1}$$

where s_c is a logit and σ the sigmoid. Findings are independent, so this is a multilabel problem. The training set splits into an expert set $\mathcal{G}$ with $|\mathcal{G}| = 58$ and binary labels $y \in \{0,1\}^{12}$, and a report set $\mathcal{W}$ with $|\mathcal{W}| = 4{,}349$ and only text. Performance is the mean of the per finding ROC AUC,

$$\mathrm{AUC}_{\text{macro}} = \frac{1}{12} \sum_{c=1}^{12} \mathrm{AUC}\big(\{y_c^{(i)}\}, \{\hat{p}_c^{(i)}\}\big). \tag{2}$$

The hidden test studies are scored by Kaggle when the submitted notebook is rerun offline. Fig. 1 shows the final system; the rest of this section explains how each part of it followed from the evidence before it.

B. Data and exploratory analysis

The training split contains 4,407 studies and 24,371 series, with 5.53 series per study on average (3 to 14). Every study contains all three planes (sagittal 40.5, coronal 35.3 and axial 24.2 percent of series), and fluid sensitivity and fat suppression flags yield six acquisition types. A series holds 33.6 slices on average, with a median of 30 and a range of 11 to 320, so slices must be ordered physically and sampled at relative rather than fixed positions.

Fig. 2 shows the middle slice of all six series of one study. The same knee looks very different across series, so a single series model discards most of the study; every model from the 2.5D network onward keeps several series and aggregates them by plane. Three report openings read *"Técnica: RMN de la rodilla. Resultados: Rotura…"* (Spanish), *"MRI of left Knee with Locator, SG PD FatSat…"* (English) and *"Antecedentes Clínicos: Síndrome meniscal rodi…"* (Spanish). Reports switch language and style freely and mix history, technique and findings in one block of text. A classifier fitted to a few dozen such reports is unlikely to cover this variety, a concern the early experiments below confirm.

Label scarcity and prevalence: The twelve label columns are empty for 4,349 studies (98.7 percent), while reports and metadata are never empty. Weak supervision is therefore unavoidable, and we treat unlabelled studies as unlabelled, never as negative. Within the 58 expert studies prevalence ranges from 60.3 percent for effusion (35 studies) to 15.5 percent for MCL injury (9 studies), and a study carries 4.1 positive findings on average. Findings cluster: effusion and synovitis are jointly positive in 22 studies. With only 9 MCL positives, any MCL AUC computed on this set is very uncertain, so we rank models by macro AUC and read per finding values cautiously (Section IV).

Reports: Reports average 147.5 words (range 8 to 690), and Spanish, Dutch, French and English all appear within the first few rows [30].

C. Preprocessing

Each series is read, ordered and normalised in the same way for every model in Branch A.

Physical slice ordering: For slice i with orientation vectors $\mathbf{r}, \mathbf{c}$ and position $\mathbf{q}_i$ we sort by

$$z_i = \mathbf{q}_i \cdot \frac{\mathbf{r} \times \mathbf{c}}{\|\mathbf{r} \times \mathbf{c}\|}, \tag{3}$$

falling back to the instance number; on a sampled series this gave a steady 4.6 mm step between neighbours.

Intensity normalisation: After applying the DICOM rescale slope and intercept and inverting MONOCHROME1 images, we estimate the 1st and 99th percentiles q_1, q_{99} over the whole series, sampled on a stride 8 pixel grid, and map

$$\tilde{x} = \frac{\min(\max(x, q_1), q_{99}) - q_1}{q_{99} - q_1}. \tag{4}$$

Normalising per series keeps neighbouring slices comparable. An early version of our cache omitted the $-q_1$ shift; fixing it was one of the changes between our 486 and 750 label runs. Fig. 3 shows a raw and a normalised slice.

Resizing and 2.5D triplets: The longer side is scaled to 224 pixels, the shorter side zero padded, and the result stored as an 8 bit image. For a series with n ordered slices and $J = 10$ positions we take centres

$$c_j = \mathrm{round}\Big(0.05(n-1) + 0.9(n-1)\tfrac{j}{J-1}\Big), \quad j = 0, \ldots, J-1, \tag{5}$$

which skips the first and last slices, as these often lie outside the joint, and stack each centre with its neighbours as colour channels,

$$T_j = \big[\, S_{c_j - 1},\ S_{c_j},\ S_{c_j + 1} \,\big] \in \mathbb{R}^{3 \times 224 \times 224}. \tag{6}$$

This gives a 2D backbone a little through plane context. We select one series per plane and then a second series per plane with a different fluid or fat suppression setting, up to six series. For the 1,808 studies used in our final Branch A run this produced 8,968 series, about five per study, and at most 60 triplets per study.

D. Early supervision strategies and their limitations

Report teachers: Our first hypothesis was that the 46 training reports could teach a text classifier to label the remaining reports. Frozen multilingual BERT embeddings with logistic regression labelled the reports, and an MRNet style AlexNet trained from scratch on the 204 most confident studies scored 0.500. A five fold KBioXLM teacher, a fine tuned biomedical multilingual encoder, labelled 954 studies for an ImageNet AlexNet and gave 0.525; the teacher reached only 0.517 OOF AUC and its labels stayed within about 0.08 of each finding's prevalence, so the image model learned little beyond the prior. This failure suggested that the bottleneck was not the text but the reliability of a labeler fitted to 46 reports.

Unlabelled images: We therefore removed the text teacher and asked whether pretrained visual features and unlabelled scans could carry the signal. A CLIP ViT B/16 [25] with adapters, trained as a multiple instance model with a Mean Teacher [26] enforcing confident consistency on unlabelled studies, beat every report teacher (0.560 with 400

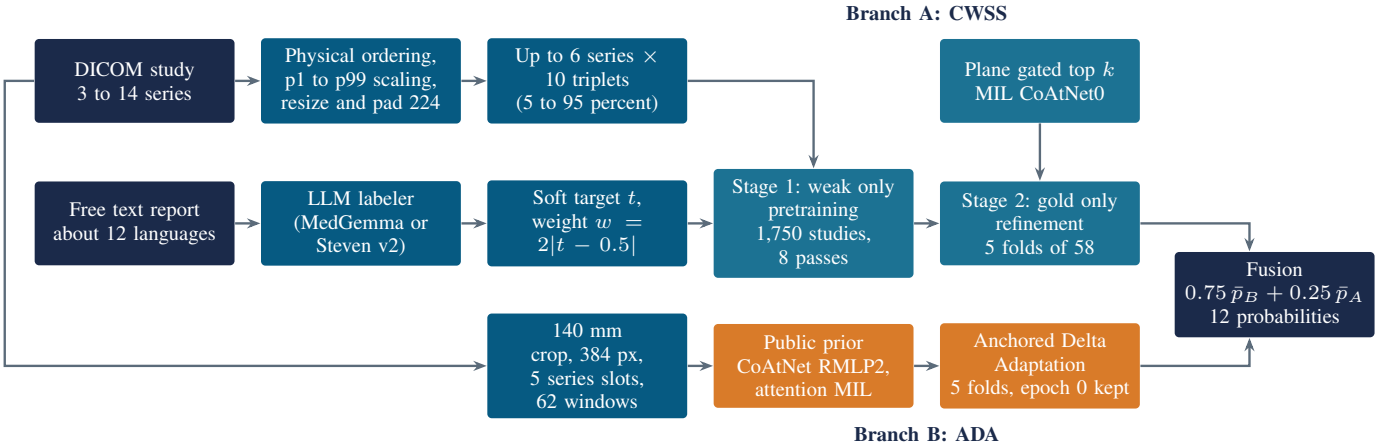


Fig. 1. End to end system. Branch A learns from LLM read reports in a weak only stage and is then refined on the 58 expert studies. Branch B adapts a public 384 pixel model to the same expert studies. Gold labels never enter Stage 1.

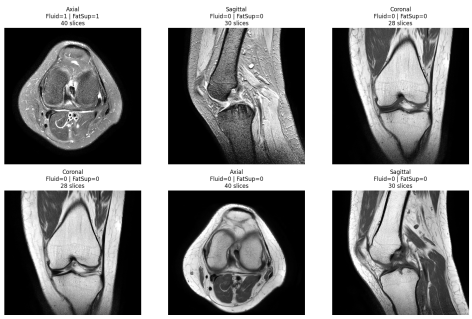


Fig. 2. Sample images: the middle slice of every series of one training study, labelled with plane, fluid sensitivity, fat suppression and slice count.

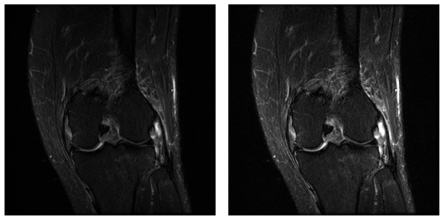


Fig. 3. Intermediate result of preprocessing: raw middle slice (left) and the same slice after series level percentile normalisation (right).

and 0.589 with 1,000 unlabelled studies) but stayed far from useful. Its fixed 12 study holdout reported 0.659, which is why later models use 5 fold OOF instead (Section IV).

E. Multi series 2.5D representation

Both early families pointed to the same need: an image model that uses the whole multi series study. Each triplet T_k of a study is encoded by an ImageNet pretrained CoAtNet0 [23], a hybrid of convolution and attention, of which only the final stage (12.9 million parameters) is trainable. A small head gives per triplet logits

$$\mathbf{z}_k = \mathbf{W}\,\mathrm{Drop}\big(\mathrm{LN}(\phi(T_k))\big) + \mathbf{b} \in \mathbb{R}^{12}. \tag{7}$$

For each plane π and finding c we average the two largest logits among that plane's triplets $\mathcal{K}_\pi$,

$$u_{\pi,c} = \tfrac{1}{2}\big(z^{(1)}_{\pi,c} + z^{(2)}_{\pi,c}\big), \tag{8}$$

where $z^{(1)}_{\pi,c} \geq z^{(2)}_{\pi,c}$ are the two largest values of $z_{k,c}$ over $k \in \mathcal{K}_\pi$. This is a softer version of the max pooling used by MRNet [1]. A learned matrix $\boldsymbol{\beta} \in \mathbb{R}^{4\times 12}$ gates the planes per finding over the set $\mathcal{A}$ of planes present in the study:

$$\alpha_{\pi,c} = \frac{\exp(\beta_{\pi,c})}{\sum_{\pi'\in\mathcal{A}} \exp(\beta_{\pi',c})}, \qquad s_c = \sum_{\pi\in\mathcal{A}} \alpha_{\pi,c}\, u_{\pi,c}. \tag{9}$$

Missing planes are masked out, so the head works for any subset of series. The top two average reflects that a lesion may appear on only a few slices, and the plane gate that some findings are clearer in particular planes.

Trained on expert labels only, this network reached 0.637 on the hidden test set with three series and six triplets, 0.644 with up to six series and eight triplets, and 0.652 from a KneeMRI pretrained backbone [9]. Better coverage helped only slightly. The architecture could learn, but 46 expert studies per fold could not supervise it: supervision, not architecture, had become the bottleneck.

F. LLM derived weak labels

Because the report teachers failed for lack of training reports, we changed the source of weak labels rather than the complexity of the classifier: an LLM can read multilingual reports without being fitted to them. We used two label sources. The first, which we produced, prompts MedGemma 27B [29] to return one of four statuses per finding, mapped to a soft target t and weight w:

$$(t, w) = \begin{cases} (0.95, 1.0) & \text{positive} \\ (0.05, 1.0) & \text{negative} \\ (0.50, 0.2) & \text{uncertain} \\ (0.50, 0.0) & \text{not mentioned.} \end{cases} \tag{10}$$

"Not mentioned" receives zero weight because a radiologist's silence is not evidence of absence, and "uncertain" a small weight because hedged language carries some, but less, information. The second is a public set of LLM report probabilities (Steven v2) [32]. For a probability $t \in [0, 1]$ we set the confidence weight to

$$w = 2\,|t - 0.5|, \tag{11}$$

so that a label at 0.5 contributes nothing and a confident label counts fully; the network learns most from statements the LLM is sure of and is not pulled towards ambiguous targets. Over the same 1,750 studies and twelve findings (21,000 targets), MedGemma gave 14,617 targets with nonzero weight (69.6 percent) and Steven v2 gave 16,942 (80.7 percent). Every label row is checked against the SHA256 hash of its report, and weak studies are asserted disjoint from expert and test studies.

Trained jointly with the expert studies in one loop, MedGemma labels lifted the network from 0.652 to 0.685 with 486 weak studies and to 0.745 with 1,750 (Table I, block C). This raised a second question. In every joint epoch thousands of noisy targets sit beside 46 expert studies, so the larger and noisier set can dominate the gradient. How should the two sources be scheduled?

G. Confidence Weighted Staged Supervision

CWSS answers that question by separating representation learning, which needs many studies, from the final decision boundary, which should be set by the trusted labels.

Stage 1, weak only: Starting from ImageNet weights, the trainable parameters are optimised only on weak studies with

$$\mathcal{L}_{\text{weak}} = \frac{\lambda}{12} \sum_{c=1}^{12} w_c \,\mathrm{BCE}\big(s_c, t_c\big), \qquad \lambda = 0.75, \tag{12}$$

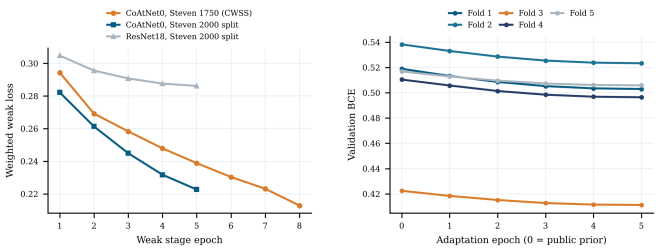
for eight passes over 1,750 studies (about 6.3 GPU hours on a T4). No expert study, label or score influences this stage, and the final epoch is kept. Fig. 4a shows the intermediate result: the weighted loss falls steadily from 0.294 to 0.213.

Stage 2, gold only: For each of five folds of $\mathcal{G}$, a copy of the Stage 1 model is refined on 46 or 47 expert studies with a fresh AdamW optimiser [27] and plain BCE,

$$\mathcal{L}_{\text{gold}} = \frac{1}{12} \sum_{c=1}^{12} \mathrm{BCE}\big(s_c, y_c\big), \tag{13}$$

for eight epochs, keeping the epoch with the lowest validation BCE. Weak studies are never read in this stage. In every fold the best epoch was the first or second, after which validation loss rose. Stage 1 therefore did most of the work, and Stage 2 behaves like a light calibration on clean labels. The test prediction of Branch A is the mean of the five fold models, $\bar{p}_A$.

With identical MedGemma labels, staging raised the hidden test score from 0.745 to 0.783, and Steven v2 confidence weights reached 0.794, our best single model; this is the main evidence for staging (Section V).



(a) (b)

Fig. 4. Intermediate results. (a) Weighted loss during weak only pretraining for CWSS and for the two models of our split 2,000 run. (b) Validation BCE of each fold during Anchored Delta Adaptation; epoch 0 is the public prior.

H. Anchored Delta Adaptation of a public prior

CWSS gave a strong model, but a stronger, task specific prior was also publicly available, raising a different question: can complementary prior knowledge be adapted to the scarce expert set without being damaged? Branch B starts from that CC0 model [31]: a CoAtNet RMLP2 at 384 pixels on a 140 mm physical crop, with five series slots of 64 slices and 62 windows of three adjacent slices. Window features $\mathbf{h}_k$ are pooled with a finding specific attention,

$$\begin{aligned} a_{k,c} &= \mathrm{softmax}_k\big(\mathbf{v}_c^{\top} \tanh(\mathbf{U}\mathbf{h}_k)\big), \\ s_c &= \mathbf{w}_c^{\top} \textstyle\sum_k a_{k,c}\, \mathbf{h}_k + b_c, \end{aligned} \tag{14}$$

in the spirit of attention based multiple instance learning [24].

ADA writes the adapted weights as $\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \boldsymbol{\Delta}$, where $\boldsymbol{\Delta}$ is nonzero only on the final backbone stage and the head, and every other parameter and every normalisation statistic stays frozen. Learning rates are 10^{-6} (backbone) and 10^{-5} (head) with cosine decay. Freezing most weights and keeping learning rates tiny limits how far 46 studies per fold can move the prior, and the unmodified prior is always a candidate, so adaptation is only kept if it helps validation:

$$e^{\star} = \arg \min_{e \in \{0,1,\ldots,5\}} \mathrm{BCE}_{\text{val}}\big(\boldsymbol{\theta}_0 + \boldsymbol{\Delta}_e\big). \tag{15}$$

Only $\boldsymbol{\Delta}_{e^\star}$ is saved, together with the SHA256 hash of $\boldsymbol{\theta}_0$, so the adapted model can never be paired with the wrong base. Fig. 4b shows the intermediate result: validation BCE fell in every fold and every epoch, and all five folds selected epoch 5.

I. End to end system and fusion

The two branches see the knee differently: Branch B a 384 pixel crop around the joint with attention pooling, Branch A the whole field of view at 224 pixels with plane gated top k pooling. Fusion is motivated by these complementary views rather than by ensembling alone. End to end (Fig. 1), a DICOM study is ordered physically, normalised, resized and padded, reduced to selected series and 2.5D triplets, scored by the five fold models of each branch, and fused into twelve probabilities as

$$\hat{p}_c = 0.75\,\frac{1}{5}\sum_{f=1}^{5} \hat{p}_c^{B,f} + 0.25\,\frac{1}{5}\sum_{f=1}^{5} \hat{p}_c^{A,f}, \tag{16}$$

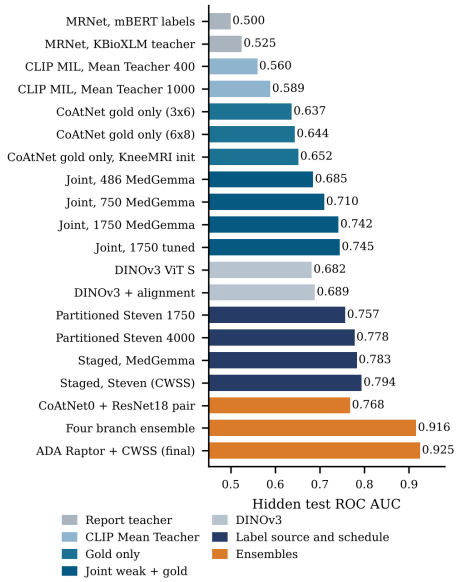


Fig. 5. Hidden test ROC AUC of every scored model, in chronological order and coloured by family.

so each Branch B fold carries 15 percent and each Branch A fold 5 percent. The weights were chosen by hand.

All models ran on Kaggle with one or two T4 GPUs and no internet at inference; Branch A uses AdamW with weight decay 10^{-4}, learning rates of 10^{-4} (backbone) and 5×10^{-4} (head), one study per batch and mixed precision.

IV. Experimental Results

A. Protocol

Each model has two numbers: macro AUC on the competition's hidden test studies, scored by Kaggle, and out of fold (OOF) macro AUC from 5 fold cross validation on the 58 expert studies, where every study is predicted by a model that never trained on it. Our earliest models used a fixed 12 study holdout instead.

B. Progression of results

Fig. 5 lists every scored model in the order we made them, and read as a whole it validates the chain of Section III. Report teachers stayed at chance (0.500, 0.525); visual semisupervision helped (0.589); a stronger representation without more labels plateaued (0.637 to 0.652); LLM weak labels moved the same network to 0.745 and staging to 0.794. A four branch ensemble with the untouched prior reached 0.916, and the final ADA plus CWSS fusion 0.925.

C. Per finding results

Fig. 6 compares OOF AUC per finding. CWSS is the strongest of our own networks on six findings and improves on the expert only network for eleven of twelve. Its weak point is MCL (0.33 OOF), with nine positives in total, so this value is close to noise. The public prior and its ADA version are far stronger, but their OOF is not untouched: the prior's author had assessed it on these studies.

TABLE I
Variation study. OOF: out of fold macro AUC on 58 expert studies (4 fold where marked †); Test: hidden test set.

Variant	OOF	Test
A. Supervision schedule (RQ1)		
Expert labels only	0.548	0.644
Joint weak and expert, MedGemma 1,750†	0.731	0.745
Staged, MedGemma 1,750 (KneeMRI init)	0.712	0.783
B. Label source and weighting (RQ2)		
Staged, Steven v2 weights (CWSS, ImageNet init)	**0.740**	**0.794**
C. Amount of weak data (joint training)		
486 studies, $\lambda = 0.2$	0.575	0.685
750 studies, $\lambda = 0.5$	0.653	0.710
1,750 studies, $\lambda = 0.5$	0.696	0.742
Partitioned across models, 1,750†	0.713	0.757
Partitioned across models, 4,000, 140 mm crop	0.702	0.778
D. Coverage and initialisation (expert only)		
3 series × 6 triplets	0.505	0.637
Up to 6 series × 8 triplets	0.548	0.644
Plus KneeMRI pretrained backbone	0.553	0.652
E. Backbone and learning paradigm		
MRNet, mBERT report labels	0.428^h	0.500
MRNet, KBioXLM report teacher	0.476^h	0.525
CLIP MIL, Mean Teacher, 1,000 unlabelled	0.659^h	0.589
DINOv3 ViT S, MedGemma 1,750	0.602	0.682
DINOv3 with cross series alignment	0.618	0.689
F. Prior adaptation (RQ3)		
Public prior, unchanged (BCE 0.501)*	0.917	n/a
ADA, epoch 5 in all folds (BCE 0.487)*	0.916	n/a
G. Fusion		
CWSS alone	0.740	0.794
50% prior, 25% CWSS, 25% CoAtNet0 and ResNet18	n/a	0.916
75% ADA prior, 25% CWSS (final)	n/a	**0.925**

^{h} fixed 12 study holdout. * prior previously assessed on these studies.

D. How far can 58 studies be trusted?

Across the thirteen models scored both ways, OOF AUC and the hidden test score agree closely (Spearman $\rho = 0.95$, Pearson $r = 0.96$; Fig. 7). The hidden test score sits 0.05 to 0.13 above OOF, as expected when OOF models train on 46 or 47 studies rather than 58. The 12 study holdout was much worse (CLIP: 0.659 there, 0.589 on the hidden test set). OOF misled us once: the staged MedGemma model had lower OOF AUC than the joint model (0.712 against 0.731) but a higher hidden test score (0.783 against 0.745), although the joint run used 4 folds and the staged run 5.

V. Ablation and Variation Study

Table I groups our models so that, as far as possible, each block changes one factor; where more than one factor changes, we say so. Only block A is a true ablation; the others are variations.

Schedule (A): Should weak and expert labels be mixed or staged? With the same 1,750 MedGemma studies and the same KneeMRI start, staging raised the hidden test score from 0.745 to 0.783. Mixing lets thousands of noisy targets dilute 46 expert studies in every epoch; staging lets the expert labels have the last word.

Labels and weighting (B): Does the label source and its weighting matter? Moving from MedGemma status weights to

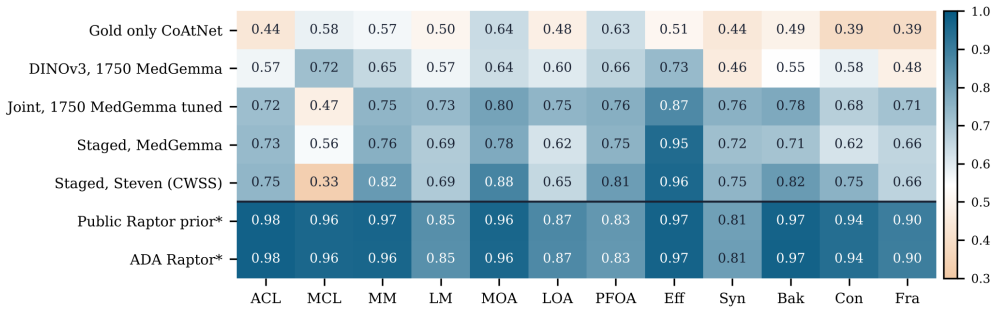


Fig. 6. Out of fold ROC AUC per finding. Rows above the line are our own networks; rows below are the public prior and its adapted version (*the prior had previously been assessed on these studies by its author). MM, LM: medial and lateral meniscus; MOA, LOA, PFOA: medial, lateral and patellofemoral osteoarthritis; Eff: effusion; Syn: synovitis; Bak: Baker's cyst; Con: contusion; Fra: fracture.

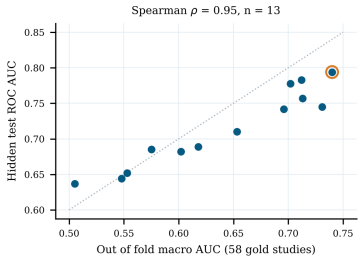


Fig. 7. Validation against the hidden test set for thirteen models. The circled point is CWSS. The dotted line marks a constant offset of 0.1.

Steven v2 confidence weights raised OOF by 0.028 and the hidden test score by 0.011. Three things changed together: the LLM, the weighting rule, which also yields 16 percent more usable targets, and the initialisation (KneeMRI for MedGemma, ImageNet for Steven v2). The comparison shows that the combination helped, not that the LLM alone did.

Amount of weak data (C): How much does weak data volume matter? More weak studies and a higher weak loss multiplier helped steadily (0.685, 0.710, 0.742, 0.745), although the multiplier changed with the study count. Splitting 4,000 studies across models still beat 1,750 on the hidden test set.

Coverage (D) and backbone (E): More series and a KneeMRI start helped a little, far less than weak labels. DINOv3 ViT S, the successor of DINOv2 [28], trailed CoAtNet0 by 0.06, and cross series alignment cut its alignment loss from 0.934 to 0.051 for a hidden test gain of only 0.007.

Prior adaptation (F) and fusion (G): ADA lowered OOF BCE by 2.7 percent with macro AUC unchanged, improving calibration without disturbing the prior's ranking. In fusion, moving from the untouched prior at 50 percent with two extra 224 pixel models to the ADA prior at 75 percent with CWSS alone raised the hidden test score from 0.916 to 0.925. Because three things changed in that step (adaptation, weights and the removed pair), we treat the gain as the effect of the whole package.

Answers to the research questions: **RQ1:** yes on the hidden test set. Staging beat mixing with identical labels (0.783 against 0.745), although OOF disagreed, possibly because of different fold counts. **RQ2:** yes, jointly. Steven v2 confidence weights beat MedGemma status weights (0.794 against 0.783), but the LLM, weighting and initialisation were not separated. **RQ3:** yes. ADA lowered validation BCE in every fold (0.501 to 0.487) without lowering macro AUC; its share of the 0.925 fusion is not isolated, and the prior had already been assessed on these studies. **RQ4:** well enough for ranking. OOF agreed with the hidden test score over thirteen models ($\rho = 0.95$) where a 12 study holdout did not, but it is optimistic and noisy for rare findings such as MCL.

VI. Limitations and Future Work

The largest share of the final score comes from a public prior that we adapted but did not design, and we have not scored it alone, so our contribution to the fused score is not isolated. The fusion weights were set by hand. OOF numbers are optimistic because each fold chose its own best epoch, the Steven v2 labels were calibrated on the expert set by their author, and the prior had been assessed on the same 58 studies. Nine MCL positives make rare finding conclusions weak, and scores on the rest of the competition's test data may differ [22].

Future work will score the prior and ADA alone, learn fusion weights from out of fold predictions, extend weak only pretraining to all 4,349 reported studies, fix the refinement epoch in advance, compare LLM labelers under one weighting rule, and apply CWSS to other musculoskeletal regions.

VII. Conclusion

Expert label scarcity was the central bottleneck: report teachers fitted to 46 reports stayed at chance, and stronger visual representations helped without solving it. LLM derived weak labels made supervision at scale possible, and their order mattered: weak only pretraining followed by expert only refinement beat joint training (RQ1), and confidence weighting helped further (RQ2), giving our best internally developed model at 0.794. An adapted public prior (RQ3) fused with CWSS reached 0.925, to be read with the limitations above,

and 5 fold OOF on 58 studies ranked models usefully (RQ4). When expert labels are scarce and reports plentiful, how weak supervision is scheduled and weighted deserves as much attention as the network that receives it.

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